CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

M.Tech. (EC)(C.S.E.) Sem. - I Examination, 2009-10
EC- 705 Advanced Digital Satellite Communication

Date: 24.12.2009, Thursday **Time: 1.00 p.m to 4.00 p.m** **Maximum Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Answer to the two sections must be written in separate answer books.**
3. **Figures to the right indicate *full* marks.**
4. **Make suitable assumptions and draw neat figures wherever required.**

SECTION- I

Q-1 (a) An uplink operates at 14 GHz, and the flux density required to saturate the transponder is -120 dB (W/m^2). The free space loss is 207 dB, and the other propagation losses amount to 2 dB. Calculate the earth station [EIRP] required for saturation, assuming clear sky conditions. Assume [RFL] is negligible. 2

(b) Describe the characteristics of geostationary satellite. 2

(c) Define & discuss the ***Keplerian elements*** in brief. 3

(d) What do you mean by ***three axis stabilization*** of satellite body? 2

(e) What do you mean by ***energy dispersal***? 2

Q 2 (a) With the help of neat diagram, justify the needs for TT & C as a subsystem for a successful satellite communication system. 4

(b) With the help of neat diagram, explain wide band receiver for satellite system. 'It is referred to as redundant receiver', justify. 4

(c) Determine the limits of visibility for an earth station situated at mean sea level, at the latitude 48.42 degrees North, and the longitude 89.26 degree West. Assume a minimum angle of elevation of 5 degrees, R = 6400 Km, a_{gso}= 42164 Km. 4

OR

Q 2 (a) Draw the block diagram of ***direct broadcast satellite system***. Explain the same for C & Ku band. 4

(b) Discuss the ***basic elements of redundant earth station*** with the help of neat block diagram. 4

(c)		For eccentric elliptical satellite orbit with an apogee and perigee points at a distance of 50000 Km and 8000 Km respectively from the center of earth. Determine the semi major axis, semi minor axis and orbit eccentricity.	4
Q 3	(a)	Define ***saturation flux density*** and prove that $[EIRP_S]_U=[\Psi_S]+[A_0]+[LOSSES]_U-[RFL]$	6
	(b)	Explain how intermodulation noise originates in a satellite link, and describe how it can be reduced. In a satellite circuit the carrier to noise ratio are: uplink 25db; downlink 20db; intermodulation 13db; Calculate the overall carrier to noise ratio.	6

OR

Q 3	(a)	Distinguish between the ***cross polarization discrimination*** and the ***polarization isolation*** phenomena.	4
	(b)	Explain various causes and effects of transmission losses occurring in satellite communication system.	5
	(c)	Define ***effective isotropic radiated power***. A transmitter feeds a power of 10 W into an antenna which has a gain of 46 db. Calculate the EIRP in i) Watts ii) dbW.	3

SECTION- II

Q 4	(a)	Enlist various frequency bands used in satellite system.	1
	(b)	Write the characteristics of Iridium satellite.	2
	(c)	Compare the uplink power requirement for Time Division Multiple Access and Frequency Division Multiple Access techniques.	3
	(d)	Define the terms ***connect clip*** and ***freeze out*** used in connection with digital speech interpolation technique used for satellite communication.	2
	(e)	Enlist the physical and orbital characteristics of advanced Tirous - N space craft.	3
Q 5	(a)	Draw the geometry to determine a sub satellite point in meridian plane and discuss the same with help of necessary formula.	4
	(b)	Enlist various propagation impairments, their causes, effects and prime importance for satellite communication system.	4
	(c)	Explain how depolarization is caused by rain. How does it differ from depolarization caused by ice?	4

OR

Q 5 (a)	Explain why a minimum of four satellites must be visible at any earth location utilizing the GPS system for position determination.	3
(b)	Determine the look angles and the range for a geostationary satellite at 30 degrees, for an earth station at latitude -20 degree, longitude -30 degree. The earth station is situated 1000m above mean sea level. (a_E=6378.1414Km, e_E=0.08182, a_{GSO}=42164.14Km, Az=78.9 deg)	6
(c)	Write the technical note on mobile satellite system and the services offered by it.	3
Q 6 (a)	Describe the channeling scheme for SPADE system for Intelsat satellite. Discuss its DASS based communication system.	5
(b)	What do mean by ***unique word***? Discuss its detection arrangement.	4
(c)	Draw and explain SPEC transmitter and receiver systems for PCM multiplexed signals.	3

OR

Q 6 (a)	Write technical note on time division multiplexing and discuss various errors associated with it.	4
(b)	Draw and explain the functional block diagram of Anik-E for C ***OR*** Ku band space craft.	4
(c)	Describe the principle of operation of Code Division Multiple Access. What effects do the unwanted signals have on the wanted signal?	4